Part 6. Implementation and Delivery

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How to open the dashboard

Open the folder `Part 6 - Implementation and Delivery/dashboard/` and double-click `index.html`, or open it in any web browser or in VS Code. No server, no command and no internet connection are needed: the data, the chart specifications and the Vega libraries are all local files in that folder. The dashboard is made for a computer screen in full-screen mode (F11): on a 16:9 screen of 1536 × 864 px or larger, each screen fits without scrolling. The main screen opens first; the link "Countries of the guests" in the header leads to the detail screen, "Accessibility help" leads to `accessibility.html`, which explains how to use both screens with the keyboard and a screen reader, and "Night mode" switches to a dark theme.

What was built

The dashboard has the two screens planned in Part 4 for Natalia, the tourism analyst of Part 2. The main screen (Figures 1 and 2) is the overview: four key figures and three charts that answer when the city hotel and the resort receive guests (Q1), when bookings are cancelled (Q5) and how the price per night changes (Q7). The detail screen (Figures 3 and 4) answers where the guests come from (Q2), when each market travels (Q3), how long it stays (Q4) and how far in advance it books (Q6). Both screens share one frame: a header band with the title, the screen links and the night-mode button, four key figures, a filter panel on the right with the active filters and a reset button, and a one-line footer with the source, the date and a button that opens the notes about the data.

After the first version was delivered, the students refined the dashboard in a separate copy and approved it before it replaced the first one. They decided that each screen must fit one 16:9 screen without scrolling, chose the Okabe–Ito colours with a night mode, and asked for more varied and more interactive charts: a calendar heatmap, a dumbbell chart, animated bubbles and box plots. Parts 4 and 5 were updated to match.

The work is split into three notebooks in `code/`. `build_dashboard.ipynb` prepares the data with pandas, builds the eight charts with Vega-Altair and writes `data.js` and `specs.js` straight into `dashboard/`. `browser_checks.ipynb` opens the pages from the folder, exactly as the professor will, and tests them in a headless browser. `final_review.ipynb` gathers every check, reviews the four checklists of the course skills and takes the screenshots in `images/`. Each notebook was saved after "Restart and run all", in that order, with no errors.

Data preparation

The dashboard never draws the 119,201 cleaned bookings of Part 1. The notebook aggregates them with pandas into six small tables, from 12 to 5,537 rows, saved in `data/processed/`. The aggregation level was chosen so that every interaction of Part 5

still works: every table keeps the hotel and the arrival year, so the filters can act on it; the monthly table keeps one row per month, so a month can be linked across charts; and the country tables keep the ten largest markets, “Other”, and every other country with at least 20 stays, which Natalia can pick from a list (the map table keeps every country). Medians and quartiles cannot be added up, so the median price and the box plots of the length of stay and the booking advance are computed in advance for each of the 12 combinations of hotel and year filter, and the bubbles for each calendar month as well. Every table was checked against the cleaned data and adds up to its totals: 119,201 bookings, 75,007 stays and 145,980 guests.

One decision changes what Natalia sees. The country charts count guests of stays that were not cancelled, not bookings, because she plans campaigns for the people who actually travel. With this measure Portugal falls from 40.7 % of the bookings to 25.8 % of the guests, since Portuguese bookings are cancelled much more often (56.7 %).

Charts

Every chart was chosen by its function in the Visual Vocabulary and built in Vega-Altair with a plain-language title phrased as Natalia’s question, a subtitle, axis titles with units, formatted tooltips, an explicit colour scale and a `description` that becomes its alternative text. The charts are drawn as SVG, never canvas, as the accessibility skill requires, and each one fills the box the page gives it, so the layout can fit the screen.

On the main screen, the guests (C1) and the price (C3) are line charts over the whole period, with the guests chart as the focal point, as the course on dashboards recommends. The cancellations (C2) are a calendar heatmap: one row per hotel and year, one column per month. The percentage of bookings cancelled is shown in seven classes of ten points (ColorBrewer Purples), not as a continuous gradient, and dots whose density grows with the class are drawn on the cells, so the value is not read from colour alone (the heatmap rule of the pattern skill); the subtitle explains the dots. On the detail screen, a dumbbell chart (C4) puts the two hotels on one row per country, so the gap between them is read by position, and a switch shows a world map (C5) in the same box instead. The bubble chart (C9) places each of the ten largest markets by its median days booked in advance and the share of its year that arrives in a month, with the size for guests; pressing Play moves the bubbles from January to December. Two box plots (C7, C8) show the spread of the length of stay and of the booking advance, because the data is skewed and a median alone would hide it.

The two hotels use the Okabe–Ito blue (#0072B2) and vermillion (#D55E00); the selected country is bluish green (#009E73) and the other countries are grey with a dark outline. With Natalia’s deuteranopia the hotel colours keep different hues but similar lightness, so each hotel also has its own line style (solid or dashed), point shape (circle or square) and a direct label: this is redundant coding (Ware). The selected country also has a light band behind its row, a bold label, a bigger outlined dot, white stripes on its boxes and an outline on the map. The night mode uses lighter versions of the same hues (#56B4E9 and #E69F00) on dark boxes and reverses the sequential scales, so “more” stays the strongest colour. Every data colour reaches 3:1 against its box in both themes. The map follows the GeoHandout: five named classes of a sequential blue scale, a TopoJSON world map at 1:110 million, drawn with the Equal Earth

projection, which, unlike Mercator, keeps the areas of the countries comparable.

Each chart was checked by code against the DRACO hard rules on its real specification: 239 checks on the eight charts, all passed (the map is outside DRACO and was judged with the GeoHandout). The values drawn by Vega were compared with the pandas tables for all 12 filter combinations, every month of the animation and a country chosen from the list: 624 marks on the main screen and 3,118 on the detail screen, all equal.

From the notebook to the pages

The pages are plain HTML, CSS and JavaScript around the Altair charts, because the keyboard navigation, focus rings and announcements required by the accessibility skill cannot be produced by Altair alone. The skills load the data as a `data.json` file, but browsers block reading a local `.json` file from a page opened by double-click. The notebook therefore writes the data as `data.js` (741 KB, including the world map) and the chart specifications as `specs.js` (51 KB), and the pages load them with `<script src>`. The data and the specifications stay out of the HTML, as the skills ask. Local copies of Vega 6.2.0, Vega-Lite 6.4.1 and vega-embed 7.1.0, with their BSD-3-Clause licences, are in `dashboard/lib/`, so the dashboard also works offline. The style and the code shared by both screens are in `dashboard.css` and `dashboard.js`, and `theme.js` sets the night mode before the page is drawn, so it never flashes white; these three files are more than the list in `architecture.md`.

The night mode keeps one set of chart specifications: the page swaps each light colour for its dark one before drawing, using a list written by the notebook. The choice is remembered in the browser for the next visit, and the page follows the computer's dark setting until Natalia chooses.

The layout follows the 16:9 grid of Part 4 and the cognitive accessibility limits: at most four key figures and three charts visible at once. On the detail screen the map shares a box with the dumbbell chart, and the two box plots, which share one country order, form one view. Each screen fits 1536 × 864 px and larger 16:9 windows without scrolling; smaller windows scroll down a little (72 px at 1366 × 768), because the page keeps a minimum height so that the charts stay readable. To fit, controls are 40 px high (radio rows and small buttons 32 px), above the 24 px minimum of WCAG 2.2 but below the 44 px first planned.

Interaction

All twelve interactions of Part 5 were implemented and follow Shneiderman's mantra. The hotel and year filters (I1, I2) update the key figures, the charts and the "Showing" line, and the link to the other screen carries them (I3); the year options name the partial years, so 2015 and 2017 are not compared with 2016 by mistake, and the reset button (I4) is always visible. Pointing at a month, or moving to it with the arrow keys, draws a rule at that month in the two line charts and an outline around its cells in the heatmap, and writes the guests, cancellations and price of that month in the "Selected month" box (I5, linked views). Tooltips (I6) give the exact value with its unit.

On the detail screen no country is selected at first, so all markets are shown alike. A click on a dot, bubble, box or country of the map selects a country in every chart, the

key figures and the “Selected country” box, and a second click, or the “No country” button, clears it (I7). A country outside the top ten can be chosen from a list; it is then added at the top of the country charts and as a bubble. Previous and Next move through the countries in the order shown. The Chart / Map switch (I8) changes the first box, the sort toggle (I9) orders the countries by guests or alphabetically, keeping “Other” last, and Play (I10) runs the year in steps of 1.5 seconds; it never starts by itself, can be paused and has a month slider. Every chart has a Table button (I11) that shows its data instead of the chart, and the night mode (I12) darkens both screens. Every change is announced to screen readers through a polite live region.

Some problems were solved on the way. Vega-Lite 6.4.1 wrote “undefined” for the hotel in the tooltip of its nearest-point selection, so the selection was moved to an invisible copy of the points. Vega did not reorder the countries when the sort order changed, so the three country charts are drawn again instead. Layers that could be empty, such as the trail of the selected country, made Vega warn about an infinite scale, so they are always drawn and hidden with opacity instead. The final checks also found that the “No country” button lost the keyboard focus when it disabled itself, that the map table stayed empty after the switch and that the filter panel overflowed by a few pixels with a country selected; the three were fixed. The browser checks test every interaction on both screens: 55 checks, all passed.

Accessibility

Each chart is one Tab stop; inside it the arrow keys move from mark to mark, Alt with the up or down arrow changes hotel, Home and End jump to the ends, and Enter and Escape show and hide the exact value next to the mark. A blue-and-white ring drawn above the chart marks the focused mark, also in the night mode, the other series are dimmed, and each move is announced with its position and value. On the main screen the focused month is linked in the three charts; on the detail screen focus only shows the value and Enter selects or clears the country, so moving through a chart never changes the selection by itself. Every chart is in a figure with a short alternative text, a long text with its main values for the current filters, and a data table. Two skip links, a visible focus on every control, `lang="en"`, valid HTML and the reduced-motion setting complete the WCAG 2.2 AA layer. The help page lists the keys, explains the night mode, the switch, the animation and the tables, and gives the course’s virtual campus as the contact for problems.

These features were tested in a browser with the keyboard only (44 checks) and the text alternatives and tables in four filter states (32 checks), all passed. The 558 visible texts of the three pages were measured in both themes: the lowest contrast is 4.58:1, above the 4.5:1 of WCAG AA, every page text is at least 16 px and every chart text at least 12 px. At 200 % zoom and at tablet widths (1024 × 768 and 768 × 1024 px) the page turns into one column, every chart is still drawn and the page never scrolls sideways.

Review and limitations

In the final review all DRACO hard rules pass, the data-viz and notebook checklists are met, and the WCAG and cognitive-accessibility checklist is met, with the adaptations

decided in `architecture.md`: `data.js` instead of `data.json`, and a dashboard for a computer or a tablet only, so the phone items do not apply. Two items are only partly met. The heatmap shows the percentage cancelled by colour classes and dots rather than by position, a choice of the students; exact values are in the tooltips, the value box and the table. On the detail screen no single chart takes 40 % of the area, as the data-viz skill suggests for a focal point: the three boxes take 22 %, 22 % and 39 %.

Some limitations remain. The data points are about 8 px, below the 24 px target size of WCAG 2.2, but every point can also be reached with the keyboard, and every country with the list and with Previous and Next. The dashboard has not been tested with a real screen reader; only the page structure, the live regions and the keyboard were checked in a headless browser. In the data, 26.8 % of the rows repeat an earlier row and are kept, so counts and cancellation rates may be too high; the key figures say "duplicates included" next to every cancellation rate. The currency of the price is not stated and euros are assumed, and the country code CN is taken as Canada; all three are in the notes about the data.

Figures

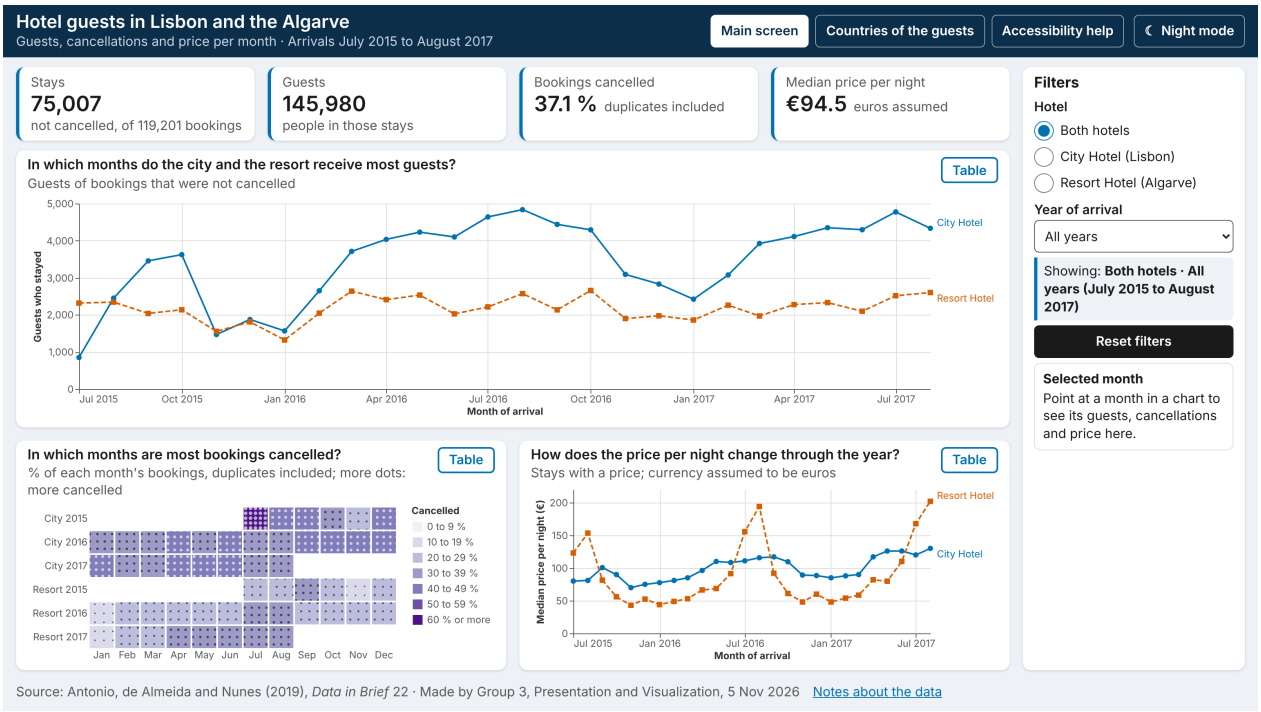


Figure 1. Main screen at first load, both hotels and all years, 1536 × 864 px. Source: dashboard/index.html; screenshot: final_review.ipynb.

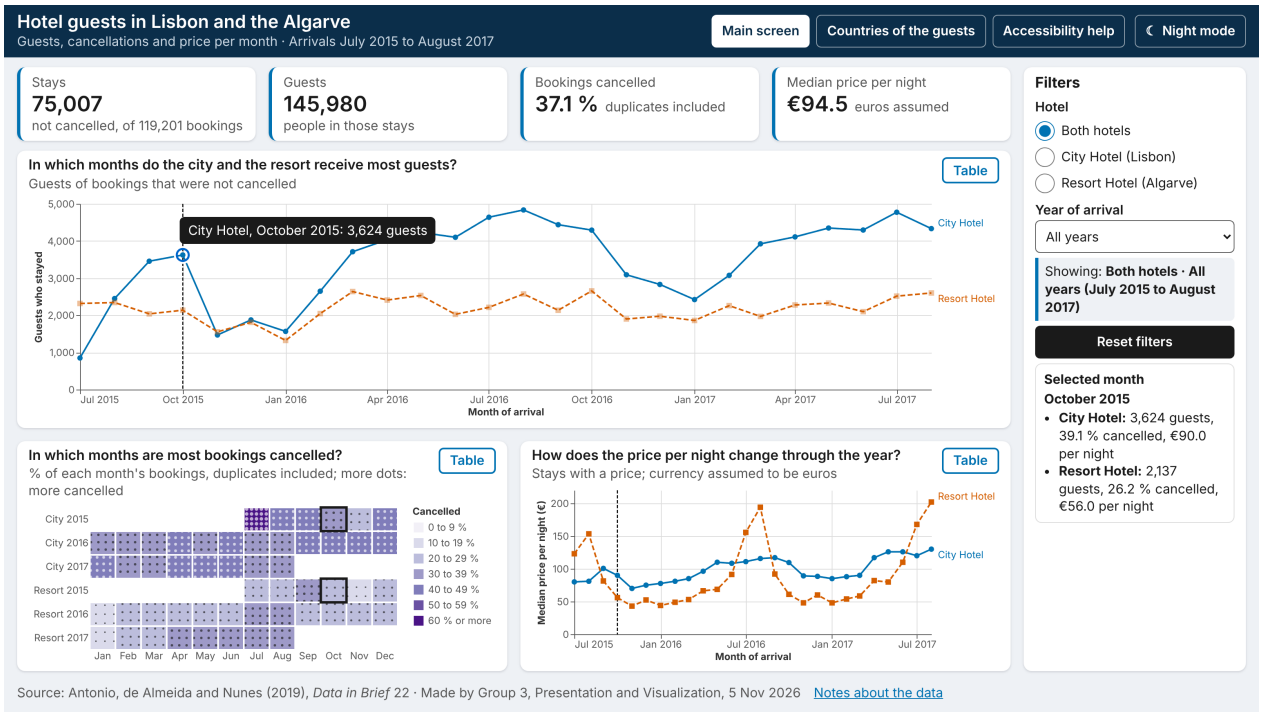


Figure 2. Main screen used with the keyboard, with the focus ring, the value box and the linked month. Source: as Figure 1.

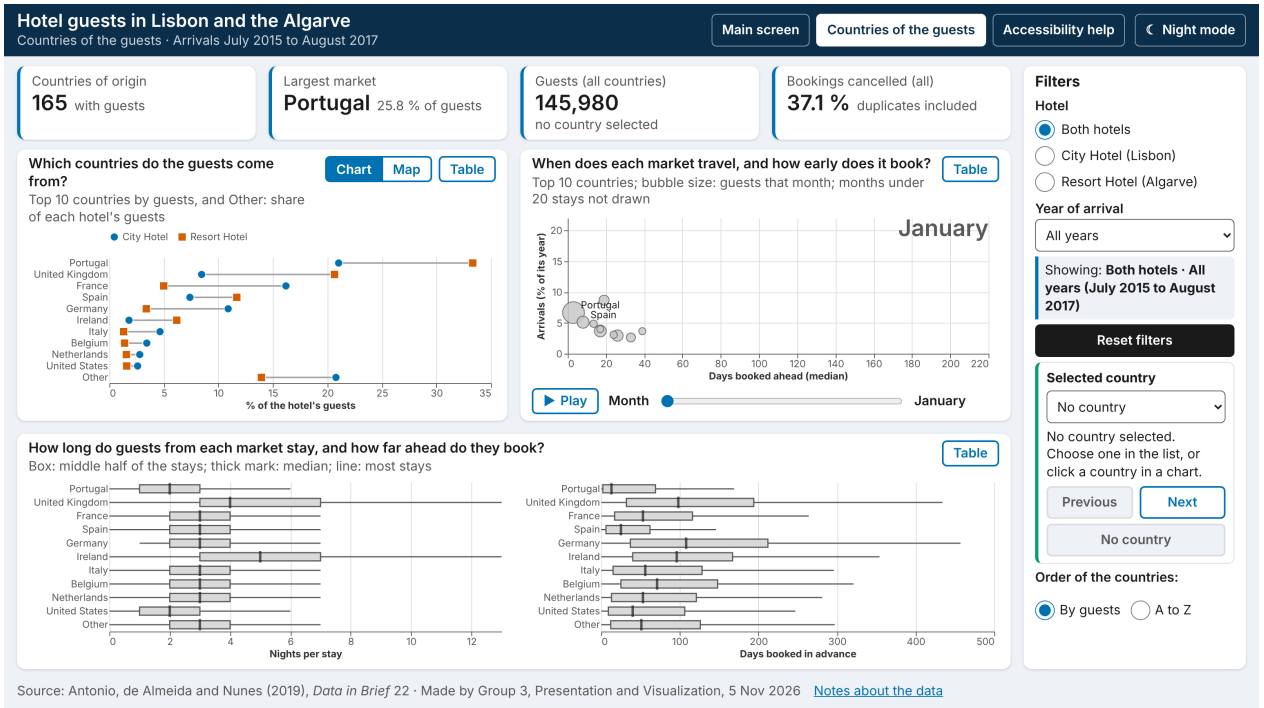


Figure 3. Detail screen at first load, with no country selected. Source: dashboard/detail.html; screenshot: final_review.ipynb.

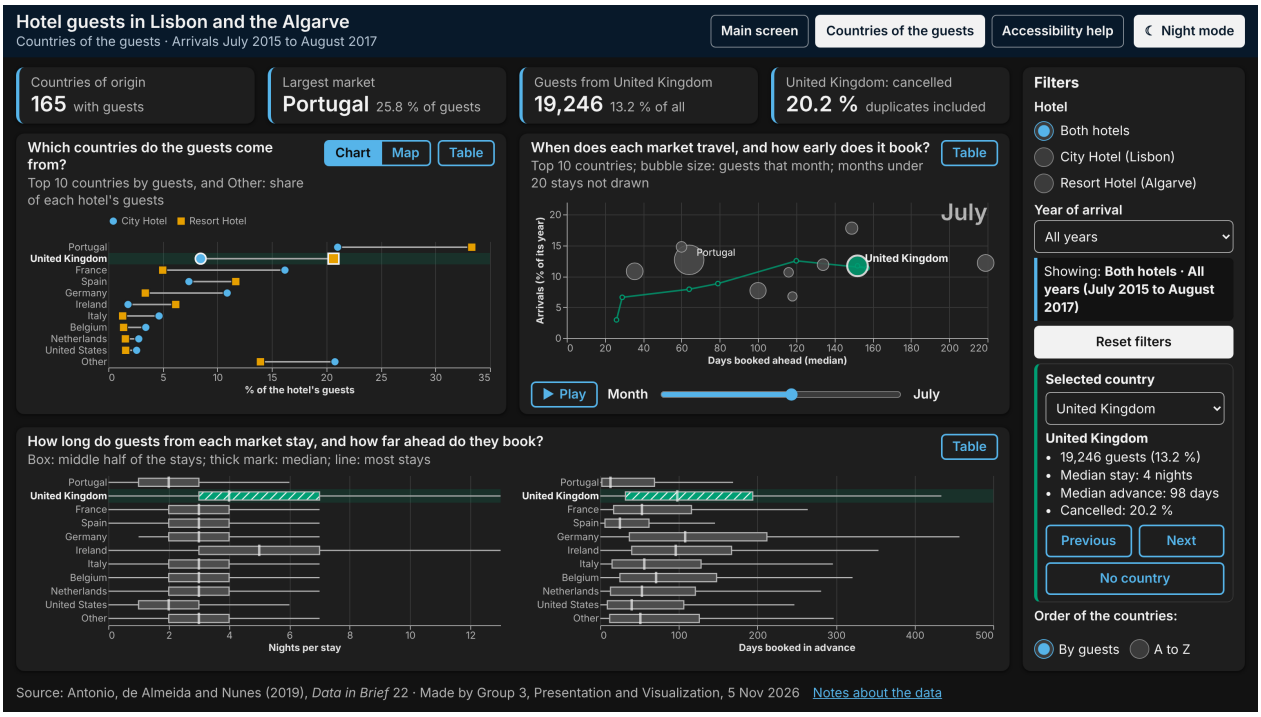


Figure 4. Detail screen in night mode, United Kingdom selected, bubbles for July. Source: as Figure 3.